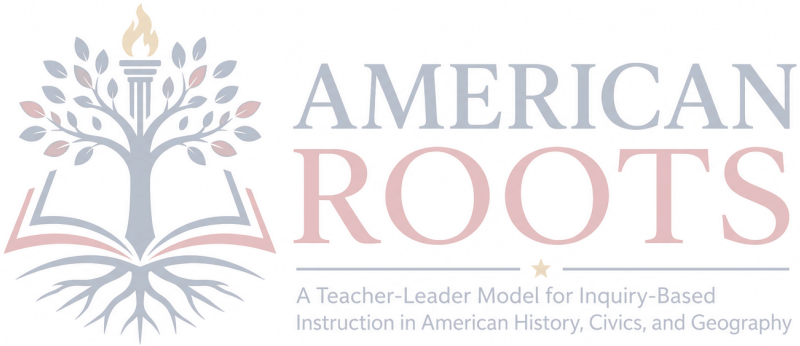
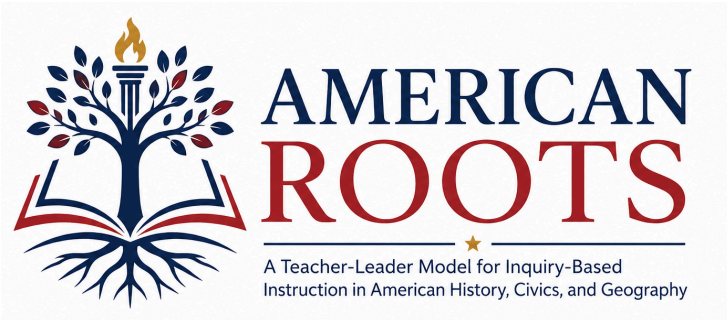
Project Narrative

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Introduction and Project Overview

As the nation marks its Semiquincentennial, America's classrooms hold a singular responsibility and opportunity, to ensure that the next generation understands the founding principles, documents, and institutions of the American constitutional republic and is prepared to sustain the experiment in self-government the founders began. Our educational institutions play a critical role in sustaining democratic engagement by equipping teachers, and in turn students, with the knowledge and skills to ensure the next generation of informed citizens. From the founding of the nation until today, education has been essential to the continuity and strengthening of our democratic system by fostering knowledge of US history and institutions, supporting the skills necessary for informed citizenship, and creating and maintaining communities of individuals working toward a common purpose. However, despite the classroom's essential role in cultivating future citizens, there has been a dearth in training and support for educators to foster this civic mission. This is reflected in the decline of civic and historical knowledge among American students. On the 2022 National Assessment of Educational Progress, eighth-grade civics scores fell for the first time since the assessment began, and history scores reached their lowest recorded level (National Assessment of Educational Progress, 2022). A central, addressable cause is the insufficient preparation and support of teachers, who frequently lack sustained, evidence-based professional development that deepens

their own knowledge of the American political tradition and equips them to teach the founding documents well to every student.

American Roots: A Teacher Leader Model for Inquiry-Based Instruction in American History, Civics, and Geography addresses this need directly with a sustained professional learning program to develop Teacher Leaders and teaching units to support educators as they teach US history, civics, and geography to 5th grade students. The University of California, Irvine (UCI) Teacher Academy proposes a Teacher Leader model that develops, implements, evaluates, and disseminates for voluntary national use a set of inquiry-based instructional units in American history, civics, and geography anchored in the founding documents and in the local places where American history is visible to students. Over a 36-month project period (October 1, 2026–September 30, 2029), the project will prepare cohorts of Teacher Leaders to author rigorous research-based units grounded in primary sources; pilot and refine those units in their own classrooms; introduce the units each summer to a broader cohort of classroom teachers through a place-based summer institute; and support those teachers through the academic year as they implement the units, collect student work, and document results.

The project pursues four intertwined goals to support teacher professional learning. First, teachers will deepen their historical content knowledge of the colonial and founding period of the United States with a focus on the Declaration of Independence and the Constitution and Bill of Rights and the historical context leading up to the creation of these foundational texts. This content knowledge development will be supported with disciplinary literacy instruction, teachers will learn evidence-based methods for guiding students in the close reading of demanding primary sources and in constructing evidence-based arguments about the American past and present. Second, teachers will learn to use AI as a pedagogically guided thinking partner for

instructional planning and curriculum development. This project will prepare teachers to use their growing content knowledge and evidence-based pedagogical understanding to provide AI with strong context, clear instructional goals, and appropriate constraints, enabling AI to support the quality and efficiency of curriculum design while preserving teachers' central role in evaluating, adapting, and guiding instruction that builds historical understanding and disciplinary literacy, consistent with the U.S. Department of Labor's Artificial Intelligence Literacy Framework (U.S. Department of Labor, 2026). Third, these 20 Teacher Leaders will work with grant program staff to develop six inquiry-based instruction units for 5th grade history, two units per year. Fourth, the Teacher Leaders will collaborate with grant program staff to present the units to Teacher Practitioners from their school districts to build a learning community centered on inquiry-based instruction on the founding documents.

Reach and deliverables. Across three annual rounds, the project will prepare 20 Teacher Leaders, train 90 classroom teachers through three summer institutes, and produce six classroom-ready, primary-source-based units (two per Year): a Declaration of Independence sequence (Year 1), a Constitution and Bill of Rights sequence (Year 2), and a place-based local-and-national history and geography sequence (Year 3). Students across our region will benefit from the *American Roots* program through teachers' instruction of the units, impacting over 1,200 students each year of the grant program who will engage with inquiry-based units centered on primary sources relevant to our nation's founding. All units, facilitator materials, and student-facing resources will be published online and shared with our network for voluntary national use, thus further impacting teachers and students in our network and across the US.

Absolute Priority: Innovative Instruction or Professional Development in American History, Civics and Government, and Geography

American Roots is designed to develop, implement, evaluate, and disseminate for voluntary use an innovative, evidence-based professional development program in American history, civics and government, and geography. The project meets both required elements of the Absolute Priority.

Improve the quality of teaching and student achievement in elementary and secondary schools. The program is built on the features of professional development that a rigorous research base associates with changed teaching practice and improved student outcomes: sustained duration, content focus, active learning, collaboration, modeling, coaching and expert support, and structured reflection (Darling-Hammond, Hyler, & Gardner, 2017; Desimone, 2009), paired with the applied classroom implementation and individual support that meta-analytic evidence links to the largest effects (Kraft, Blazar, & Hogan, 2018). Teachers deepen content knowledge with historians and engage in reading and discussing primary sources; develop pedagogical content knowledge for teaching founding-era texts; co-author and pilot units in their own classrooms; and receive support as they implement and study student work. The federal performance measure, demonstrated gains on pre- and post-assessments of understanding of American history, civics and government, and geography, will be used to measure the impact of the *American Roots* program.

Innovation, scalability, accountability, and a focus on underserved populations. The project is *innovative* in three respects: a **Teacher Leader authorship** model in which classroom teachers, supported by content and pedagogy experts, produce and refine primary-source units rather than receiving off-the-shelf curriculum; a **place-based design** that uses local Southern

California history and the founding sites of Philadelphia, Washington, D.C., and the Eastern Shore of Maryland to make the geography of American history concrete; and a **human-first approach to AI** that teaches educators to deploy generative AI to support teacher instructional planning and curriculum development. It is *scalable* because every unit, facilitator guide, and assessment is designed for transfer and published online for voluntary national use, because the **Teacher Leader structure builds local capacity that is sustained** beyond the grant period; and because the project will provide open access models and prompts that help teachers use AI to support their own local work to prepare quality instruction and curriculum. It innovates by creating research-based and discipline-specific professional development and enduring **publicly accessible resources** that will guide educators in the use of AI to extend, expand, and adapt these materials for their own context and utilize AI to leverage the pedagogical and contextual knowledge of teachers to create additional materials. It is *accountable* through a mixed-methods evaluation with objective pre/post measures, rubric-scored unit and student-work artifacts, and implementation documentation. Its ***focus on underserved populations*** is reflected in the partnerships with local high-need districts in which large majorities of students qualify for free or reduced-price meals (see Appendix D for letters of support), and the units are designed so that students who are still developing reading proficiency, including English Learners, can access grade-level founding documents through texts that are amplified rather than simplified (Martin & Bunch, 2025).

Competitive Preference Priority 1: Promoting Patriotic Education

The project is designed to introduce participants to the founding documents and primary sources of the American political tradition in a manner consistent with the principles of a patriotic education: an accurate and honest presentation of American history grounded in

analysis of America's founding documents, the principles that shaped the founding, and the ways in which those ideals continue to influence Americans' civic aspirations today. The six units place the founding documents and primary sources at their center to support students to develop the content knowledge and skills necessary for informed civic engagement.

Founding documents and first principles. Curriculum Units developed as part of the grant program (see Appendix A, Logic Model Goal 3) will include a focus on the Declaration of Independence (Year 1) and the Constitution and the Bill of Rights (Year 2) (Priority a). Teachers and their students will understand the historical context in which the founding principles emerged and their development over time. Instruction will situate these texts within the broader American political tradition, including the influence of antecedent ideas and institutions on the founders' understanding of liberty and self-government (Priority d). Students will be taught how to engage in close reading and reasoned discussion that supports their understanding of these texts as part of America's civic inheritance and as enduring sources for reflection on the rights and responsibilities of citizenship. At the center of this learning are the historical documents that have shaped our constitutional republic. The units will provide rich instruction in reading and writing, essential literacy skills for an informed citizenry, and cultivate the classroom contexts for students to deepen their understanding of their rights and responsibilities as American citizens.

History and geography through place. Year 3 units are place-based, connecting the national founding to the local landscape students inhabit and to the founding-era geography of Philadelphia, Washington DC and the Eastern Shore of Maryland, which the Teacher Leaders will study firsthand. The founding cities are treated as designed civic landscapes—the deliberate plan of the federal city, Philadelphia's grid—so that students learn to "read" the geography of self-government in the built environment (Priority b,c). Teachers will then see the way the new

nation was organized architecturally to reflect founding values in Washington DC. Finally, the trip includes a view into the rural colonial US to understand how the earliest national settlements flourished as tobacco and grain farms on the Eastern Shore of Maryland. Local Southern California history extends the same analysis to the communities students inhabit, giving the geography topic concrete, place-based expression. This design gives the geography strand of the priority concrete expression so students learn to read the American story in monuments, archives, civic institutions, and the built and natural environment of their own communities, early rural America, and of the founding cities.

Response to Competitive Preference Priority 2: Returning Education to the States

The project is designed to be carried out in coordination with California's own educator-development infrastructure and to channel federal investment through State-endorsed structures rather than constructing a parallel federal program. The UCI History Project and California Reading and Literature Project at UCI are both part of the California Subject Matter Project (CSMP), a statewide network administered by the University of California Office of the President under longstanding State statutory authority (California Education Code section 99200 et seq.) and operated in partnership with the California Department of Education. The project has received the endorsement of the State Superintendent of Public Instruction, meeting Competitive Preference Priority 2 (See Appendix D for letters of support).

Quality of Project Design

Project Rationale

The *American Roots* program will directly support 110 fifth-grade teachers in increasing and improving the quality of their instruction in history, civics, and geography, while developing six classroom-tested inquiry-based units for use by a national audience of educators, and

resources that help teachers use AI as a pedagogically-guided support for instruction and curriculum development. To build local capacity and ensure the units are classroom-tested and scalable, *American Roots* will develop the leadership capacity of 20 Teacher Leaders from partner districts as they deepen their content knowledge, strengthen their implementation of disciplinary literacy practices, and design, implement, and refine curricular units that center the founding documents. These units will serve as the foundation for three summer institutes for 90 teacher practitioners, extending the program's impact on fifth-grade US history instruction. Together, these activities address a pressing need for professional learning and high-quality curricular resources that support inquiry-based instruction in elementary history, civics, and geography.

The problem the grant addresses is well documented: students' civic and historical knowledge has declined (National Assessment of Educational Progress, 2022), and the proximate, addressable lever is the teacher and the resources they use for teaching. Teachers' own knowledge of the founding documents and their pedagogical capacity to teach demanding primary sources to every student must be enhanced. A national survey of elementary teachers highlights that professional learning in history and civics was "underprovided," with teachers lacking access and support to engage in disciplinary professional development. In addition, this challenge is compounded by a lack of resources to support rigorous instruction (Diliberti, Melissa, et al, 2023). The civic stakes are equally well established by longitudinal evidence indicating that students who experience high-quality civic and historical instruction show higher subsequent civic knowledge, engagement, and efficacy, across socioeconomic backgrounds (Hess & McAvoy, 2015; Kahne & Sporte, 2008).

The project's theory of action follows from this rationale. *If* Teacher Leaders deepen their content knowledge of the founding documents and their pedagogical content knowledge for teaching primary sources and constructing evidence-based arguments, and author, pilot and refine inquiry-based units in their own classrooms, using AI as a guided support for curriculum design and revision; and *if* those units are introduced each summer to a broader cohort of teachers who implement them with support during the academic year; *then* participating teachers'content knowledge and instruction will improve, and their students will develop deeper content knowledge and disciplinary literacy skills in American history, civics and government, and geography, necessary for civic engagement.

The design also reflects the program's emphasis on innovation and sustainable implementation. Rather than delivering a fixed curriculum, the project develops local capacity through a Teacher Leader authorship model and produces transferable, openly-licensed online units and facilitator materials so that educators elsewhere can adopt them. Professional development and open access resources on research-based use of generative AI by teachers for history instruction allows teachers to design, adapt, refine, and extend the curriculum to meet the demands of their context, enabling them to better reach and teach all students. The place-based and human-first-AI elements are deliberate innovations responsive to the present moment. The Semiquincentennial invites place-based study of the founding and its impact on communities across America, while the rapid diffusion of generative AI underscores the need to prepare teachers to use AI with pedagogical judgment: to contextualize materials, refine instructional supports, and strengthen curriculum design without displacing the teacher's central role in guiding student learning (Bastani et al., 2025).

Logic Model and Project Design

Our approach will center on effective, evidence-based professional development that is sustained over time, grounded in disciplinary content, and connected to teachers' instructional practice. The project will bring together experts in history and civics education, professional learning, curriculum design, disciplinary literacy, and AI-supported instruction to design and implement a sustained professional development program. The program will center history and civics content, engage educators as leaders in creating teaching resources tied to their classroom contexts, and create structured opportunities for critical thinking, feedback, reflection, and revision. These features intentionally align with elements of professional development associated with changes in teaching practices and positive student outcomes including sustained duration, content focus, active learning, collaboration, and expert support (Darling-Hammond, et.al. 2017).

The *American Roots* program will create sustained opportunities for educators to deepen their historical content knowledge and strengthen their use of disciplinary literacy practices, so that they can support their students to engage in meaningful inquiry, evidence-based reasoning, and projects that foster civic engagement. This approach reflects a deeper learning pedagogy in which teachers move beyond coverage of material and make informed instructional decisions about what students understand, where gaps remain, and what experiences and adaptations will help students reach shared learning goals (Darling-Hammond, K. & Darling-Hammond, L. 2022). Professional learning on generative AI will further support this work by helping teachers use AI in pedagogically sound ways to design, adapt, refine, and extend curriculum while keeping their professional judgment at the center. Because deeper learning requires teachers to inquire into their own practice, collect data from students, and revise instruction over time, *American Roots* centers Teacher Leaders as curriculum developers. Through cycles of co-design,

classroom implementation, feedback, and revision, the program will strengthen teachers' instruction and support students' development of the content knowledge, disciplinary skills, and civic understanding needed for informed participation (Theory of Change, Image 1, and Logic Model, Image 2).

Image 1: Theory of Change

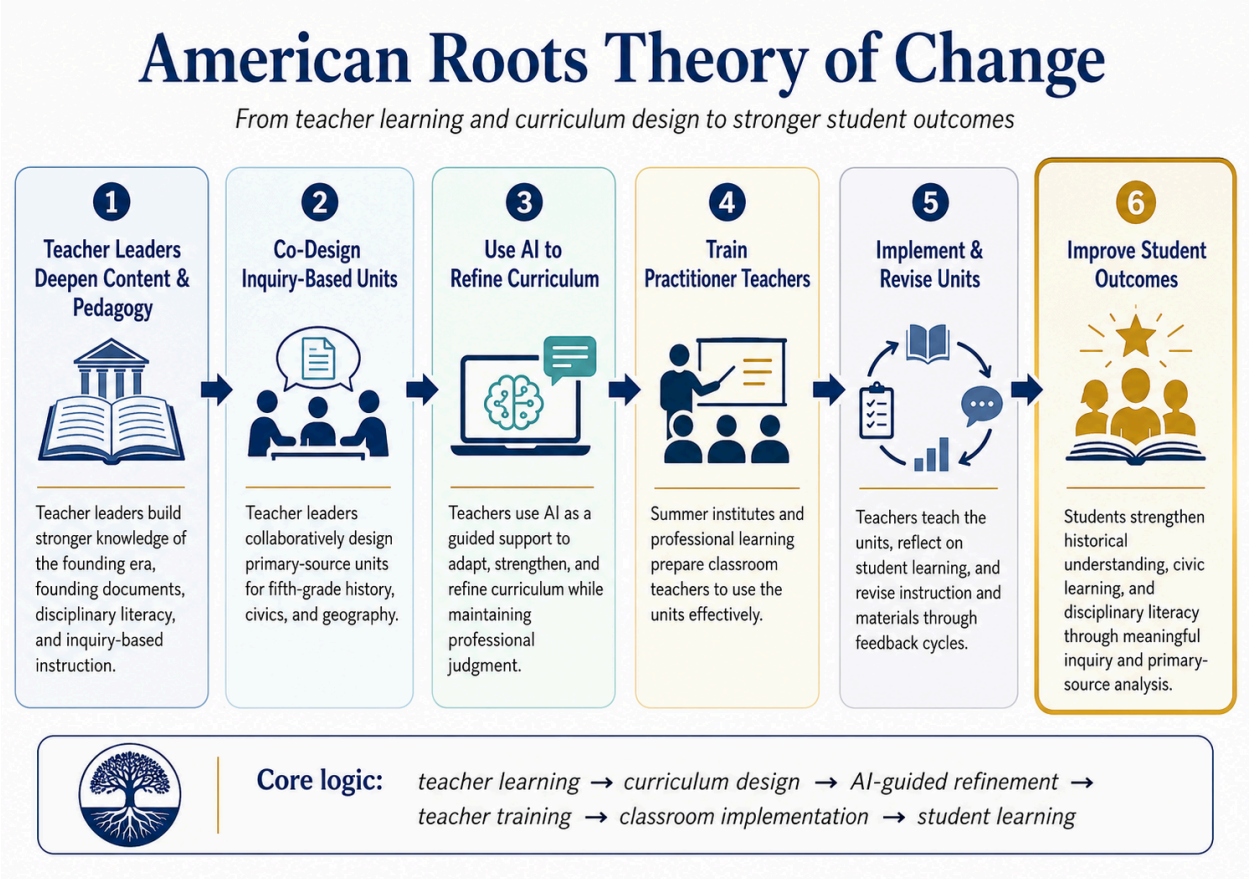
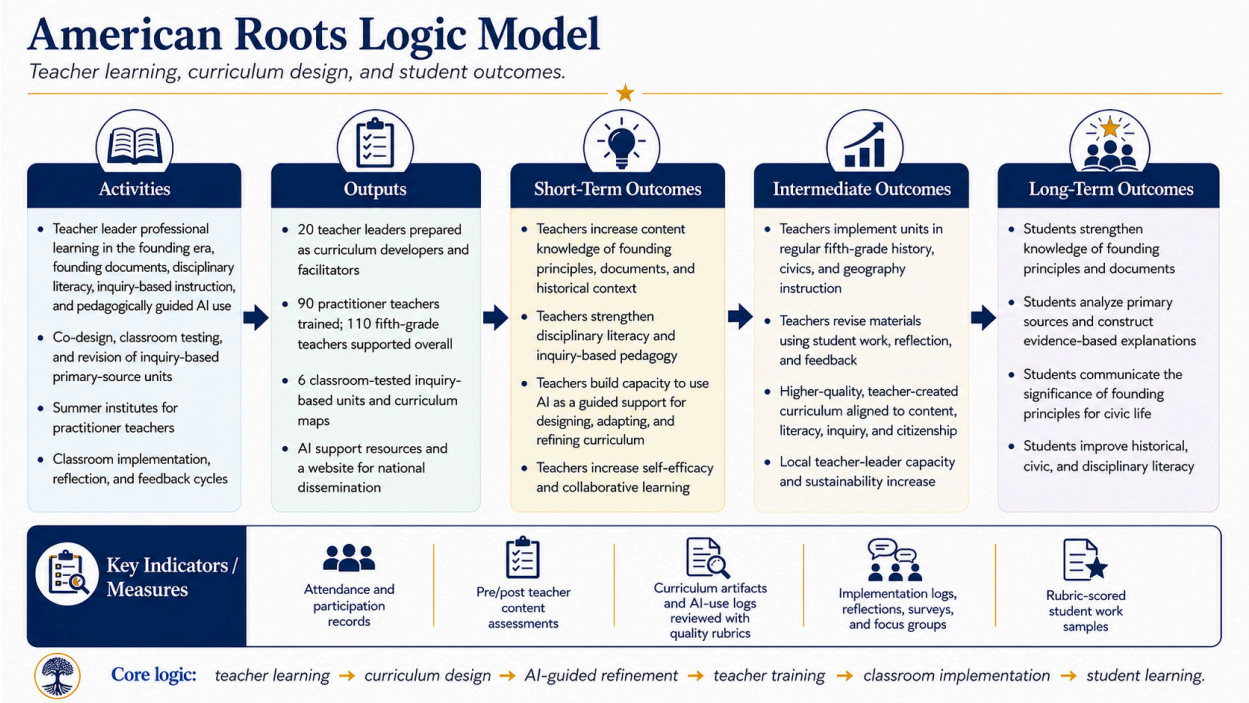


Image 2: Logic Model (The full Logic Model can be found in Appendix A)



Goal 1: Increase teacher content knowledge of founding principles and documents

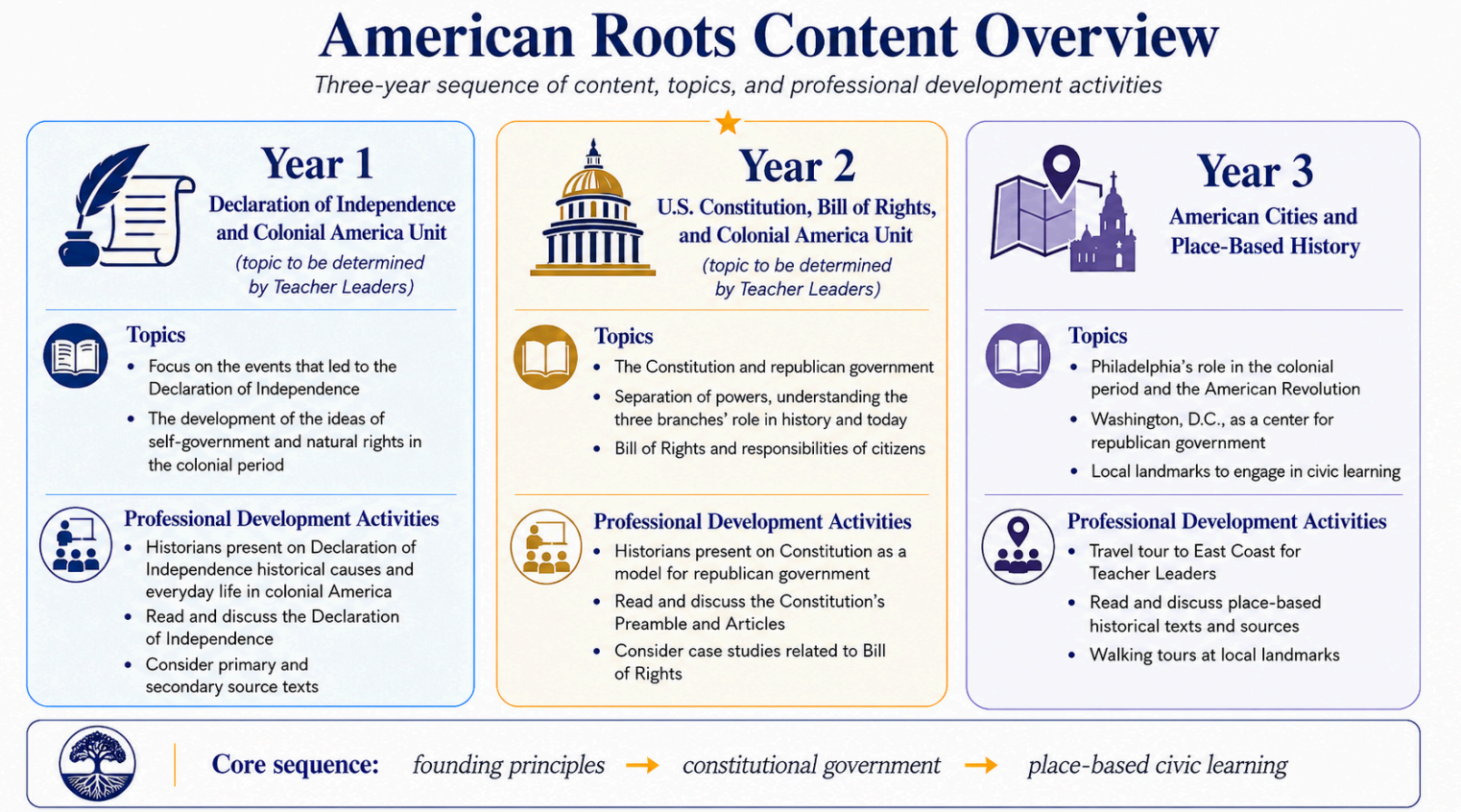
The *American Roots* program will center on the study and commemoration of the founding documents of the United States to support teachers and their students to increase their content knowledge in US history, civics, and geography. Each year will include a focus on specific founding documents, the **Declaration of Independence** (Year 1) and **US Constitution and Bill of Rights** (Year 2), that play a significant role in understanding American constitutional democracy past and present, and help prepare students to fully participate in their communities in informed ways. Through this in-depth study of the American political tradition, teachers will examine the ideas, interpretations, and enduring influence of the founding texts on American life. Historians will provide essential context for understanding how the people, ideas, and institutions of the founding era emerged and evolved, and how they continue to shape Americans' rights,

responsibilities, and participation in civic life. Teachers in the program will examine archival artifacts that reflect the civic and political debates of the founding era and consider how to integrate these primary sources into classroom instruction. Teachers will explore collections such as those in the National Archives and Library of Congress to identify primary sources that illuminate the principles of liberty and self-government in American history and support inquiry-based instruction in fifth-grade classrooms. This will include multi-media sources such as literature, maps, artifacts, and other significant documents to support students to consider the many ways Americans have debated and demonstrated their understanding of the founding principles. Pedagogical instruction will prepare teachers to support students in the close reading and analysis of founding texts, including the Declaration of Independence, Constitution and Bill of Rights. Through inquiry-based instruction, students will examine the principles expressed in these documents, consider their continuing significance in American civic life, and connect that learning to their own responsibilities as members of their communities.

Historians and domain experts will lead teacher content knowledge development through close reading and discussion of the founding documents, with explicit attention to the instructional choices that make those texts accessible, meaningful, and learnable by fifth-grade students. The design is grounded in pedagogical content knowledge (Shulman, 1986, 1987) with the assumption that knowing history is necessary but not sufficient, teachers must construct tasks and activities that let students do the discipline's reasoning. This work of translation of historians' methods into classrooms requires deliberate pedagogical design that turns documents into objects of structured inquiry (Bain, 2005). The project gives teachers time and support (through experts and AI) to perform that translation and treats teachers themselves as a source of

expertise through the authorship model. The image below (Image 3) shows the structure and sequence of our three year-program.

Image 3: Program Structure and Sequence



An essential component of supporting students' engagement in an inquiry-based understanding of the past is their ability to access the founding documents that are demanding texts — lexically dense, syntactically complex, and drawn from a world distant from students' own (Reisman & Wineburg, 2012). The project develops teachers' capacity to teach disciplinary literacy and the social, semiotic, and cognitive practices of historians (Fang, 2012), including sourcing, contextualization, and corroboration (Wineburg, 1991; Seixas, 1993). Instruction is modeled through cognitive apprenticeship, which makes expert thinking visible and provides models of the types of historical analyses students are working toward (Collins, Brown, &

Holum, 1991). Texts are amplified rather than simplified so that students still developing proficiency, including English Learners, can access grade-level founding documents (Martin & Bunch, 2025). The same disciplinary practices extend to geographic sources. Founding-era maps available through the National Archives and the Library of Congress are themselves primary sources with authors, purposes, and points of view; teachers learn to have students source, contextualize, and corroborate them alongside textual evidence. Reading maps as arguments—about territory, settlement, and the spatial reach of authority—develops geographic reasoning within the close-reading apprenticeship rather than as a separate exercise.

Literature runs alongside these documentary and geographic sources as a parallel strand of inquiry. Novels, read-alouds, and other literary works connected to each year's focus give students a human entry point into the era and its ideas. Where a founding document states a principle, literature lets students encounter how people understood, debated, and lived it — the texture of everyday life, the weight of the choices individuals faced, and the range of experiences present in a given time and place. Research with upper-elementary students finds that well-chosen historical fiction can convey historical content as effectively as expository text, while engaging students through narrative (Dwyer & Martin-Chang, 2023). Realizing that potential depends on teachers' instructional choices: studies of fifth-grade classrooms show that when teachers integrate literature into social studies, they select quality texts purposefully to serve clear content goals (Brugar & Whitlock, 2019). Teachers in the program will learn to select and sequence literature that enriches the documentary record and to integrate it so that literary, historical, and geographic sources work together within a coherent arc of inquiry.

Goal 2: Build teachers' capacity to use AI as a pedagogically guided support for designing, adapting, and refining curriculum on the founding documents.

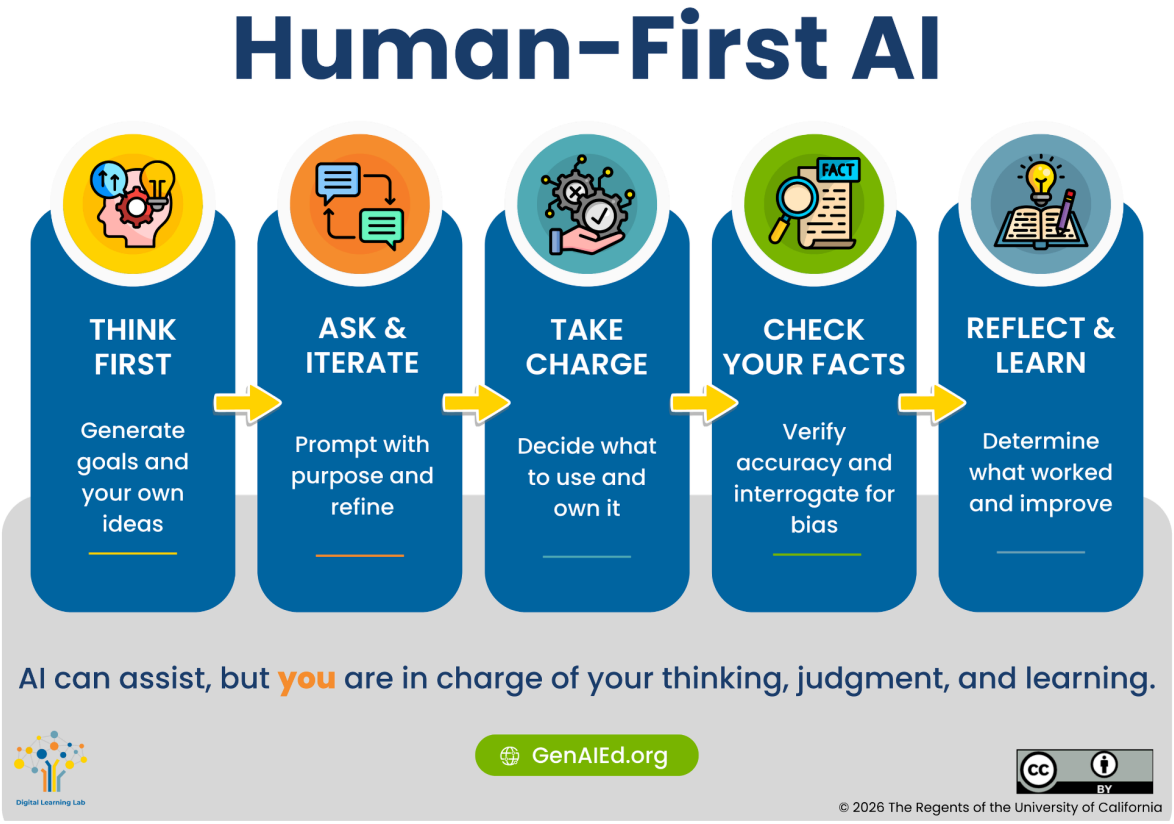
The project will build teachers' capacity to use AI as a pedagogically guided support for designing, adapting, refining, and extending curriculum on the founding documents. The central premise is that AI can strengthen curriculum development when teachers direct it with strong content knowledge, clear instructional goals, and an understanding of students' needs, interests, and developmental readiness. The teacher—not the tool—remains the author, evaluator, and instructional decision-maker.

This human-first approach reflects emerging evidence that the educational consequences of generative AI depend not simply on whether AI is used, but on how it is used. Uncritical reliance can produce the appearance of quality while weakening the judgment that produces high-quality work; disciplined use, guided by expertise, can extend professional capacity (Bastani et al., 2025). *American Roots* addresses this risk by keeping pedagogical judgment in teachers' hands. Teachers will learn to direct AI effectively, evaluate its outputs against disciplinary and instructional standards, and revise AI-supported materials before they reach students. Professional learning will focus on practical, curriculum-centered uses of AI, such as drafting and differentiating primary-source activities, generating models or item banks for teacher review, adapting materials for fifth-grade learners, and analyzing student work to identify actionable instructional next steps. AI-generated materials will not be treated as ready-to-use curriculum; teachers will be trained to verify historical claims, evaluate alignment to disciplinary and instructional standards, and revise all AI-supported materials before classroom use.

The AI pedagogy will rely on the Digital Learning Lab's Human-First AI Framework (Tate et al., 2026; Image 4) and align with the U.S. Department of Labor's Artificial Intelligence

Literacy Framework (U.S. Department of Labor, 2026) and with the technological pedagogical content knowledge the work requires (Mishra & Koehler, 2006). AI-supported curriculum work will be evaluated against the project's CLIC design criteria (discussed in detail below)—Content, Literacy, Inquiry, and Citizenship—so that AI use remains anchored in historical accuracy, primary-source analysis, disciplinary literacy, inquiry-based learning, and informed civic participation.

Image 4. Human-First AI



Where appropriate, teachers will use PapyrusAI, a UCI-developed platform that lets them interact with a large language model in a structured, scaffolded "walled garden" governed by research-based, pedagogically sound prompts, without requiring personal accounts or exposing

interactions to model training. The research team will also have access to any AI interactions on the platform for analysis, as described below in the Evaluation section. PapyrusAI has been studied in prior research on AI-supported writing and feedback (Tate et al., 2026).

Goal 3: Develop 6 inquiry-based, primary source units to support history, civics, and geography in 5th grade classrooms

Each academic year Teacher Leaders, identified and recruited by participating districts, will develop, implement, evaluate, and disseminate a set of inquiry-based instructional units in American history, civics, and geography anchored in the founding documents, culminating in six inquiry-based primary source units for 5th grade that will be available online for open-access educator use. The Teacher Leader team will be split into two groups, Teacher Leader Writers and Curriculum Pilot Teachers. Teacher Leader Writers will collaborate with program staff to develop the units during the academic year. Each year, two units will be developed, the topics will be selected based on the need for rigorous standards-aligned curriculum to support student content knowledge and disciplinary literacy. The Curriculum Pilot Teachers will implement the units in their classrooms. All Teacher Leaders will deepen their content knowledge by learning from historians and partnering with disciplinary pedagogical experts to understand how to best support students to read, discuss, and write about primary sources. After looking at student work from the pilot classrooms and considering the results of classroom implementation, the units will be revised and refined to better meet the learning needs of the students.

The units will be guided by the California History–Social Science Framework (California Department of Education, 2017) and follow a four-part design the project terms **"CLIC"**: **Content** (the founding documents and their development), **Literacy** (disciplinary reading and

writing), **Inquiry** (investigation organized around compelling questions and primary-source evidence), and **Citizenship** (informed participation grounded in the American political tradition). Each unit will be anchored by an essential question, sequence primary sources and literature from accessible entry points to more complex texts, and culminate in a student product in which students construct an argument about the founding principles and their legacy. Geographic inquiry is woven through the Content and Inquiry elements and students will analyze primary source maps. In the Declaration of Indepeendence unit, students examine the spatial dimensions of colonial grievance, including frontier policy and the geography of trade and territory. In the Constitution and Bill of Rights unit, students analyze federalism as the spatial distribution of authority among localities, States, and the national government—the geographic architecture of the constitutional republic. In the place-based unit, students study human-environment interaction and sense of place in their own communities and in the founding cities and Eastern countryside. Each unit is grounded in the geographic skills articulated in the California History–Social Science Framework (California Department of Education, 2017).

Because teachers often limit instructional time on history to favor English Language Arts, disciplinary literacy and literary texts will be integrated into the units to foster humanities instruction. Each unit pairs founding documents with literary and primary texts chosen to illuminate its core concepts. In the Declaration of Independence unit, students might read excerpts from the Declaration alongside a biography of one of its authors or signers and a work of historical fiction set in the Revolutionary era, using the story to consider how ordinary people experienced the ideas of liberty and self-government that the document asserts. In the Constitution and Bill of Rights unit, students might pair the text of an amendment with a speech, letter, or work of fiction that shows a citizen exercising or contesting that right, making abstract

guarantees concrete. In the place-based California unit, students might read local histories, oral histories, or fiction tied to their own region to connect founding principles to the civic life of their communities. Across all three units, students practice close reading, vocabulary development, discussion, and evidence-based writing while building historical, geographic, and civic understanding.

Each summer, 30 Teacher Practitioners, who teach fifth grade in partner and local school districts, will attend a four-day summer institute focused on two inquiry-based, primary source units. In preparation for the summer institute, teachers will receive instruction from historians to build the content knowledge needed to teach the units with accuracy and depth. Teacher Leaders, along with grant program staff, will guide Teacher Practitioners in hands-on practice with the units and support planning for classroom implementation. This will include opportunities for them to engage in the unit as learners so they can understand how students will read, discuss, and respond to primary sources. Teacher Practitioners will have structured time to consider how to implement this learning into their own classroom and have the opportunity to discuss with colleagues pacing, scaffolding, assessment, and other practical aspects of implementation. Teacher Practitioners will then implement the units the following academic year and will engage in follow-up meetings with program staff to discuss implementation and analyze student work to improve their own instruction to better support student learning.

Goal 4: Build a sustainable Teacher Leader network that supports rigorous inquiry-based instruction in the founding documents

The *American Roots* program is deliberately designed to develop teacher efficacy and inspire and reinvigorate teachers, combating teacher burnout, a significant challenge facing American education today. The program will embed community-building at every stage of its

professional learning model. From the cohort-based Teacher Leader preparation to the annual summer institute to the academic-year follow-up support, the program staff has structured each touchpoint to ensure that fifth grade teachers do not work alone. Instead of facing the challenges of rigorous inquiry-based instruction in isolation, participants enter a network of peers with shared goals, curriculum, and understanding about the ways that they intend to impact student learning. This intentional community design treats belonging not as a byproduct of the work but as a prerequisite for it.

The structure of the *American Roots* program reflects what Hargreaves (2019) describes as authentic collaborative professionalism, the kind of deep, teacher-driven, trust-based collaboration that research has shown to increase both teacher self-efficacy and student achievement. The *American Roots* program avoids "contrived collegiality," the top-down, administratively imposed collaboration that breeds resentment rather than renewal, by centering teacher voice, teacher authorship, and teacher leadership throughout. Teacher Leaders are not just recipients of content; they are co-authors of the instructional units, pilots in their own classrooms, and mentors to the wider cohort. This model of collaborative professionalism in which teachers define and develop their own purposes as a community, generates the kind of professional investment that sustains teachers through the inevitable difficulties of implementing rigorous new instruction.

The program's sustained, multi-year architecture is also a critical structural choice in combating teacher burnout and building lasting capacity. Darling-Hammond, Hyler, and Gardner (2017) found across 35 rigorous studies that effective professional development must be of sustained duration, engaging teachers in learning over weeks, months, or even academic years rather than in short one-off workshops. The *American Roots* model provides sustained time and

structure for shared learning and implementation. Teacher Leaders will experience continuous cycles of learning, authoring, piloting, and refining, while Teacher Practitioners move from the summer institute into school-year coaching and implementation support. Academic-year follow-up sessions give teachers recurring, structured time to return to the work together. For example, Teacher Practitioners will report on their implementation of the units, bring student work samples and lessons to review with the team and their cohort, and use what they see to refine instruction and revise units before teaching them again. This cycle of inquiry, reflection, and revision against real student work is what sustains learning across the year rather than letting it dissipate after the summer institute.

Finally, the Teacher Leader model at the heart of *American Roots* is a powerful and research-aligned strategy for cultivating the sense of inspiration, professional identity, and belonging that serves as a buffer against burnout. When teachers are elevated as leaders, authors, and mentors rather than passive recipients of externally imposed methods, they develop what Hargreaves and Fullan (2012) call professional capital, the human, decisional, and social capital that deepens with experience and collaborative investment over time. The program staff has recognized that inspiring teachers to teach rigorous American history and civics to fifth graders is not only a matter of content knowledge or instructional strategy, it is also a matter of helping educators see themselves as professionals with expertise worth sharing, ideas worth publishing, and a community worth belonging to. When teachers experience that sense of meaning and recognition, research shows they are more likely to remain in the classroom, continue refining their practice, and pass their enthusiasm on to students (Van Dusen, R & Otero, V, 2014). *American Roots* is thus not simply a curriculum project, it is a professional renewal program built on the evidence that sustainability begins with belonging to a shared learning community.

Significance

The outcomes of the *American Roots* project are significant in both magnitude and kind because of its focus on elementary educators. Improvements in teachers' content knowledge, capacity to teach primary sources, and AI literacy are consequential precisely because they compound, a Teacher Leader who masters the founding documents and authors a strong unit affects students every year thereafter, and the cohort and dissemination structure multiplies that effect across classrooms and, through open release of curricular units and resources, across the nation. Finally, the use of generative AI amplifies teachers' ability to author, adapt, refine, and extend high-quality instruction for every student. The project will prepare approximately **20 Teacher Leaders**, train approximately **90 Teacher Practitioners**, and produce **six classroom-ready units** anchored in the founding documents to foster close reading, evidence-based argumentation, and civic engagement among approximately **1,200 fifth-grade students** per year. Because participating teachers serve in high-need elementary schools, the units reach substantial numbers of students from low-income backgrounds. The contribution to teaching practice is durable, the teacher-leader model builds local expertise and community belonging that persists beyond the grant, and openly licensed units and facilitator materials extend the contribution to educators throughout the US.

The project is likely to produce meaningful improvement measured against rigorous and relevant standards. Units are aligned to the California History–Social Science Framework (California Department of Education, 2017) and to the disciplinary practices of sourcing, contextualization, corroboration, and evidence-based argumentation that define rigorous history and civics instruction (Wineburg et al., 2011). Teacher knowledge gains are measured against performance measures through pre/post assessments; unit quality is judged against an explicit

rubric; and student learning is assessed through work products scored with co-developed rubrics aligned to State content and literacy standards. The likelihood of improvement is supported by the evidence base for the project's components (Darling-Hammond et al., 2017; Kraft et al., 2018) and by the prior results of the applicant's professional-development programs for history teachers in Southern California, discussed next.

The *American Roots* program Principal Investigator has past experience leading professional learning for history educators that demonstrated a positive impact on teacher and student learning. English-Learner (EL) students were positively affected by their teachers' participation in the UCI History Project's Improving Teacher Quality grant, with EL treatment students demonstrating higher scores than comparison students in historical thinking skills, as determined by the outside evaluator. Additionally, the program team has expertise crafting both content and disciplinary literacy assessments for both teachers and students. The program staff will develop the assessment tools in Winter 2027 alongside historians as they develop the content for the institute programming. The assessment tool will include multiple choice content questions and short answer questions aligning to the content topic of the summer institutes (2027, 2028, 2029). These assessments will measure the growth in teachers' content knowledge as a result of their learning from content presentations by experts, opportunities to engage in content area reading, and discussions with colleagues about content implementation. An increase in teachers' content knowledge will improve instruction in the elementary classroom and improve students' knowledge of US history content and source analysis as evidenced by evaluations of our prior professional development programs. For example, students of teachers involved in UCI History Project Teaching American History programs increased their knowledge of US history content, writing skills, and primary source analysis (S. Swigert, 2006, 2007 &

Continuing Improvement Associates, 2009). Through repeated opportunities to study the founding documents and discuss how to translate that content into classroom instruction, teachers will deepen their understanding of the founding ideals and strengthen their ability to help students connect those ideals to history, civics and geography.

Each of the participants in the *American Roots* program will provide feedback after every professional learning opportunity and provide feedback on the impact of the program. We will use this survey data to inform our upcoming work to meet the needs of the teachers and develop programming that best meets their specific needs. These results will be compiled into a report to inform our team and stakeholders.

Our grant program will also assess teachers' implementation of curricular units to support student knowledge of founding documents. At each of the follow-up sessions during the academic year, teachers will report on their implementation of units and will collect and share student work samples (see Appendix G, implementation log sample). Teachers will report on the literacy strategies they implemented and bring examples of lessons and student work to review and discuss, in order to develop and deepen their understanding of students' specific needs and successes in disciplinary literacy across both documentary and literary texts.

To make this review systematic rather than impressionistic, teachers will analyze student work using the California Reading and Literature Project's (CRLP) Student Work Analysis Protocol (see Appendix G), a shared seven-step process. Teachers (1) review the rubric and success criteria for the task; (2) describe the student work with a focus on content understanding; (3) describe the language students used to express that understanding — vocabulary, sentence structures, and discourse patterns — and how those choices meet, approach, or stretch toward grade-level language arts standards; (4) interpret the work to identify what students can do and

where they struggle; (5) draw implications for instruction; (6) commit to specific instructional next steps; and (7) reflect on the collaboration itself. This focus on language alongside content draws on CRLP's established approach to supporting English Learners and keeps their needs central to instructional decision-making.

The grant program staff and Teacher Leaders will co-create rubrics to measure student responses to reading and analysis questions across the units' documentary and literary sources, including the analysis of primary sources. These rubrics will align to the Common Core State Standards for English Language Arts and the California Content Standards for History–Social Science. As part of the role of the Teacher Practitioners, we will allocate time to review and discuss student work in order to continually inform and improve instruction in students' disciplinary reading and to address student needs in unit revisions. Centering these opportunities for educators to engage in inquiry, reflection, and revision based on student data will allow for more targeted and meaningful instruction in disciplinary literacy and support the skills students need for informed citizenship. Teachers will also share successes with fellow *American Roots* teachers and colleagues in their schools as a learning community to further student achievement.

To further impact educators, the *American Roots* program's six units, as well as all related teaching materials and AI resources, will be available online and open-access (Goal 4). These materials will be available to our network of educators at our website, https://teacheracademy.uci.edu.

Management Plan

The project pursues four goals, each with specific, measurable objectives that are ambitious yet achievable within the 36-month period and aligned with the purposes of the grant program.

Goal 1. Increase teacher content knowledge of founding principles and documents

- Objective 1.1. Participating Teacher Leaders and Teacher Practitioners demonstrate statistically and practically meaningful gains from pre- to post-assessment on a validated measure of content knowledge of the founding principles and documents.
- Objective 1.2. Teachers implement instruction that reflects their learning on the founding principles and documents

Goal 2. Build teachers' capacity to use AI as a pedagogically guided support for designing, adapting, and refining curriculum and instructional materials on the founding documents.

- Objective 2.1. Teacher Leaders and Teacher Practitioners demonstrate growth in Human-First AI use, as evidenced by reflections, AI-use logs, and artifacts showing iterative prompting, verification of AI outputs, and teacher revision.
- Objective 2.2. Teacher Leaders and Teacher Practitioners use AI to design, adapt, or refine at least one instructional activity, assessment, or student-work analysis protocol aligned to the CLIC criteria.
- Objective 2.3. AI-supported curriculum artifacts meet quality thresholds on the project rubric for content accuracy, primary-source use, inquiry design, disciplinary literacy supports, accessibility for fifth-grade learners, and teacher reflection.

Goal 3. Develop 6 inquiry-based, primary source units to support history, civics, and geography in 5th grade classrooms

- Objective 3.1. By the end of each year, Teacher Leaders complete two CLIC-designed units aligned to the California History–Social Science Framework that meet a quality threshold on the project's unit-review rubric.

- Objective 3.2. Each unit is implemented in classrooms by Curriculum Pilot Teachers, revised based on implementation evidence and student work, and released for voluntary national use with facilitator materials.
- Objective 3.3 Teacher Practitioners implement two units during the academic year and submit implementation documentation and student work samples.

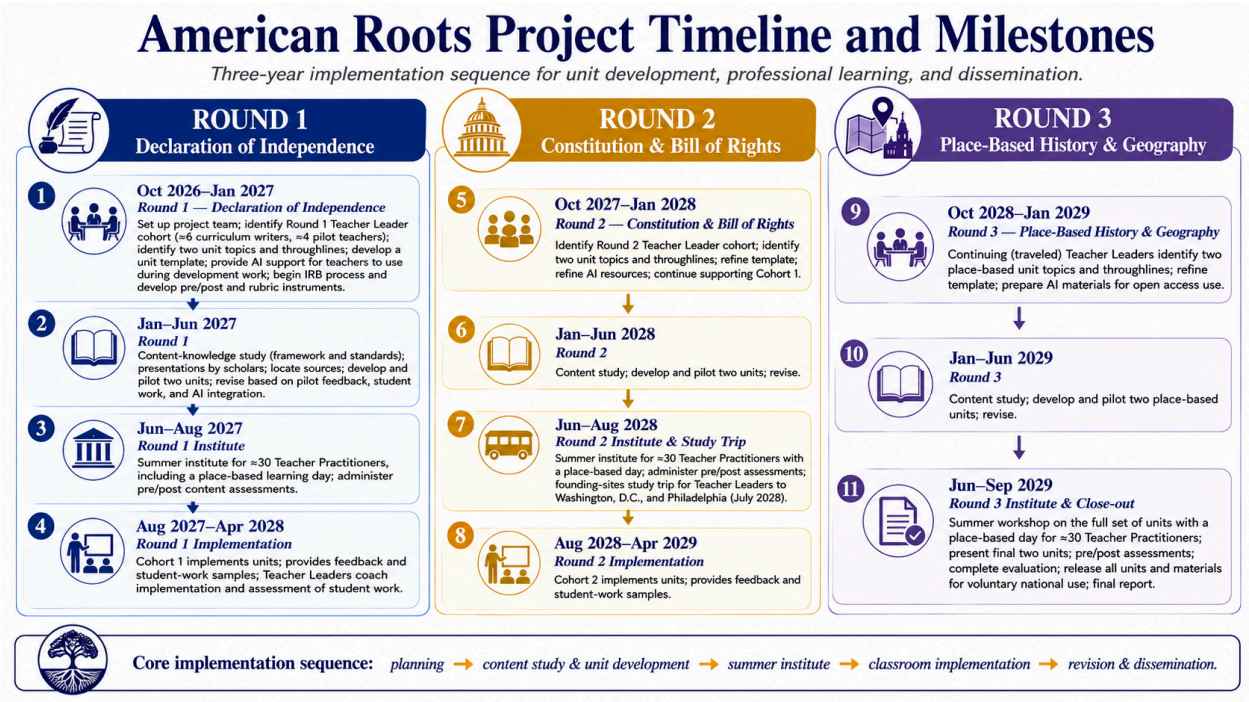
Goal 4. Build a sustainable Teacher Leader network that supports rigorous inquiry-based instruction in the founding documents

- Objective 4.1. Teacher Leaders and Teacher Practitioners engage in shared learning to implement units in their classrooms where they select teaching artifacts and student work to discuss and reflect upon to engage in evidence-based discussion for continuous improvement as measured by attendance and self-efficacy surveys.
- Objective 4.2. Teacher Leaders and Teacher Practitioners integrate the units into their regular curriculum as demonstrated by implementation reports and focus groups.

Timeline

The management plan is feasible and structured to deliver objectives on time and within budget. The project operates in three annual rounds with a repeating, well-rehearsed cycle: fall topic identification and content study; winter–spring unit development and piloting; a summer place-based institute introducing the units to the teacher practitioners; and academic-year implementation support. The repeating structure is itself a feasibility asset, each round refines a known process. In addition, the applicant team has executed comparable multi-year, federally funded teacher-development programs. Image 5 presents the timeline and milestones; responsibilities follow (see also Appendix A, Timeline and Milestones).

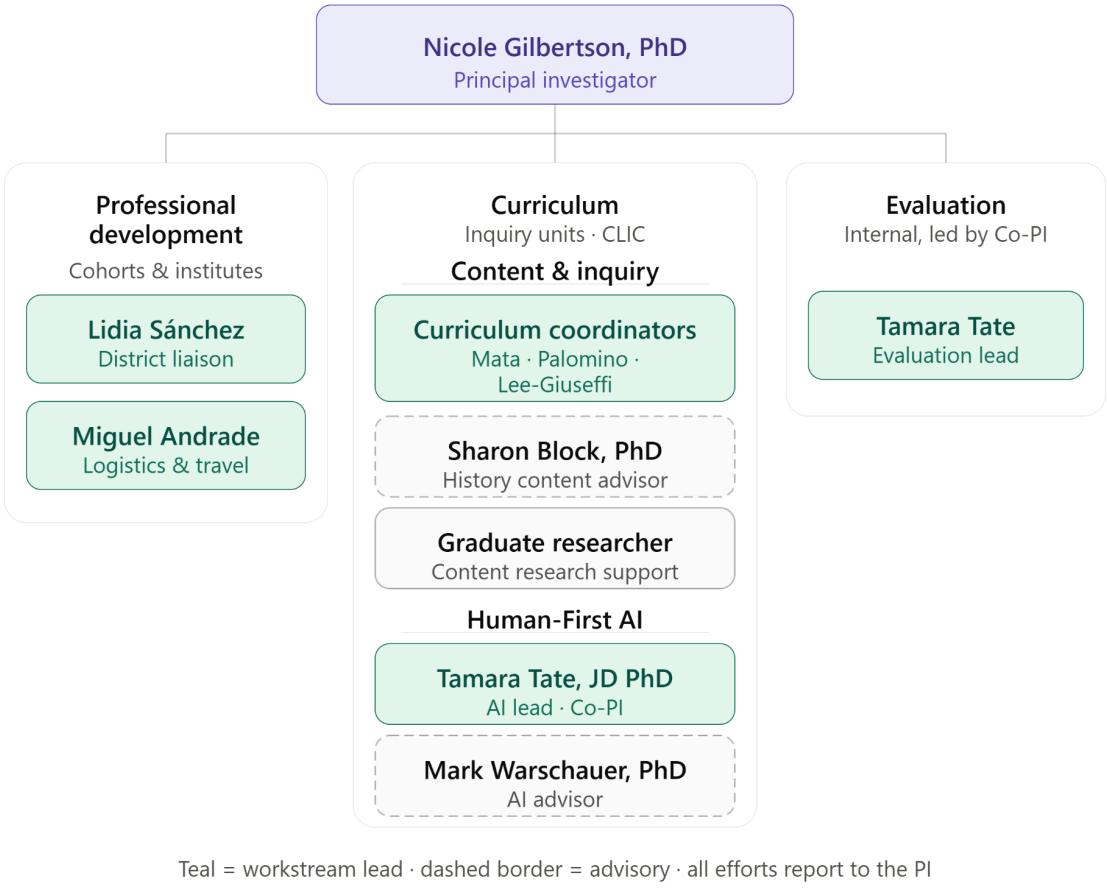
Image 5: Timeline and Milestones



Responsibilities and Coordination

Dr. Nicole Gilbertson, Principal Investigator, provides overall direction for the *American Roots* program and supervises a team whose members have led and collaborated on multiple multi-year, grant-funded teacher professional learning programs together. She works directly with the Co-Principal Investigator, Dr. Tamara Tate; the History Faculty Advisor, Dr. Sharon Block; the AI and Digital Learning Advisor, Dr. Mark Warschauer; the three Grant Curriculum Coordinators, Cindy Mata, Linda Palomino, and Robyn Lee-Giuseffi; the Grant Program Coordinator, Lidia Sánchez; a Graduate Student Researcher; and the Administrative Coordinator, Miguel Andrade. Reporting lines and functional responsibilities are summarized in Image 6.

Image 6: Reporting Lines and Responsibilities



Coordination proceeds through a regular cycle of Grant Program Team meetings that surface emerging requirements, insights, and participant feedback. Project materials are maintained on a shared, version-controlled Google Drive organized by folder and subfolder; running agendas and notes are kept in Google Docs to ensure comprehensive, accessible documentation; Zoom supports synchronous meetings; and email supports ongoing communication. Because the team has managed prior grant projects and district partnerships together, the project builds on established methods of collaboration.Continuous improvement is embedded in the design. Participant survey data and structured review of teacher-created lessons,

using a program-designed protocol, inform decisions and revisions in each subsequent round, so that each cohort benefits from evidence gathered in the round before it.

Key Personnel and Responsibilities

Nicole Gilbertson, Ph.D., Principal Investigator (25% FTE), directs all aspects of the *American Roots* program. She has led long-term federal grant programs—including a current National Activities grant, Teaching American History grants, and Improving Teacher Quality programs for Southern California history teachers—and holds a Ph.D. in history with expertise in disciplinary literacy developed in part through the IES-funded national WRITE Center, where she co-designed professional development to strengthen writing in secondary history classrooms. Dr. Gilbertson co-develops programming and assessments, authors all reporting, serves as the primary liaison to federal program officers, and oversees dissemination and publication of project artifacts.

Tamara Tate, J.D., Ph.D., Co-Principal Investigator (20% FTE), is a Project Scientist at UC Irvine and Associate Director of the Digital Learning Lab. She has served as PI or Co-PI on four grants focused on AI-supported teaching and learning, including the NSF-funded project that produced PapyrusAI, and studies technology-supported instruction and AI-supported teaching and learning. As a WRITE Center researcher, she analyzed secondary students' history writing, developed measures of teacher knowledge, and modeled relationships among teacher self-efficacy, instruction, and student achievement using hierarchical linear models. Tate has also led multiple evaluations of classroom instruction, including several RCTs, and has a strong background in both qualitative and quantitative analyses. For *American Roots*, Dr. Tate leads both the AI curriculum development and the program evaluation.

Mark Warschauer, Ph.D. (.5% FTE), Distinguished Professor of Education at UCI with appointments in Informatics, Psychological Science, and Language Science, is among the most widely cited scholars on digital and AI-based writing and a member of the National Academy of Education. He has served as PI on more than $30 million in funded research, including the IES-funded WRITE Center, and has studied AI in education for over fifteen years, most recently through three NSF-funded grants developing AI-based interactive learning media for young children. He leads the lab that developed PapyrusAI and advises the project on integrating digital tools effectively and ethically to strengthen disciplinary literacy.

Sharon Block, Ph.D. (.5% FTE), Professor of History in the UCI School of Humanities, is an expert in colonial and Revolutionary-era history, holds multiple NEH awards, including serving as an official consultant for a 250th Celebration grant, and is currently working with descendants of founding families to document their ties to the Revolutionary era. Widely recognized as the first scholar to apply large language models to historical research, she brings domain expertise on the use of AI in historical teaching and has partnered with Dr. Gilbertson on prior federal Teaching American History grants. As History Faculty Advisor, Dr. Block works with the PI and Grant Curriculum Coordinators to build teachers' content knowledge, identify primary and secondary sources, and review curricular units; drawing on her experience as a former Associate Vice Chancellor, she partners with Dr. Gilbertson to develop content assessments that measure teacher learning.

Cindy Mata (20% FTE), Grant Curriculum Coordinator, is a former middle and high school history teacher and Director of the UCI History Project, which connects the UCI History Department with K–12 history–social science teachers across Southern California and serves

educators from credential candidates to veterans. She has helped multiple districts build systemic approaches to the State Seal of Civic Engagement. Ms. Mata grounds Teacher Leaders in the fifth-grade History–Social Science Content Standards and the four instructional shifts that anchor the CLIC framework, introduces the K–8 Civic Learning Pathway to the State Seal of Civic Engagement, co-develops units, models analysis tools for primary and secondary sources, and helps prepare Teacher Leaders to present to the summer cohort.

Linda Palomino (20% FTE), Grant Curriculum Coordinator, is a fifth-grade teacher in Santa Ana Unified and Co-Director of the UC Irvine/Orange County region of the California Reading and Literature Project (CRLP), where she has designed and facilitated literacy and English language development professional learning since 2007. With more than three decades in literacy instruction and curriculum design—including work as a District Curriculum Specialist for English Learner Programs—and as a lead author of the CRLP resource *Learning How English Works*, she brings both statewide professional-learning experience and current classroom practice to the project. Ms. Palomino integrates disciplinary literacy and English language development into the units, ensures founding-era texts are accessible to English Learners, co-develops units, helps design the student-work analysis process, and prepares Teacher Leaders to facilitate the summer cohort.

Robyn Lee-Giuseffi (20% FTE), Grant Curriculum Coordinator, is a fifth-grade teacher in Santa Ana Unified and Regional Co-Director of the UC Irvine/Orange County region of CRLP, with more than two decades designing literacy and ELD professional learning. For over twenty years she served as a District Curriculum Specialist, leading elementary history–social science materials review, piloting, and implementation paired with language and literacy routines, and

she co-authored *Learning How English Works*. Ms. Lee-Giuseffi works with Teacher Leaders to build units that integrate California history–social science content with literacy and ELD instruction, co-develops units, supports academic-year implementation and follow-up, and helps prepare Teacher Leaders for the summer institute.

Lidia E. Sánchez (10% FTE), Grant Program Coordinator, is a former elementary dual-language teacher and Co-Director of the UCI CRLP. She works with the PI and partner districts to identify the Teacher Leaders and Teacher Practitioners who attend the summer institutes, to elicit district priorities for fifth-grade history, and to integrate those priorities into the units developed by the Curriculum Writers. She also coordinates with Mr. Andrade to support district reimbursement for substitute teachers.

Graduate Student Researcher (50% FTE academic year; 100% for two summer months) supports the historical research underlying unit development—conducting archival research, co-developing content presentations, and curating primary and secondary sources for Teacher Leaders and cohort teachers—and partners with grant staff to develop and analyze assessments and to prepare reports that inform programming.

Miguel Andrade (20% FTE), Administrative Coordinator, manages logistics, meeting space, virtual-meeting facilitation, participant registration and stipends, travel, and website development, and works with the fiscal office on purchasing and budgetary commitments.

Resources

The project draws on substantial institutional resources that ensure adequate support. The University of California, Irvine is a top-ranked public research university. The School of

Education has recognized strengths in language and literacy development and in digital learning. The **UCI Teacher Academy** integrates the campus's teacher professional-development organizations, including the UCI History Project and UCI California Reading and Literature Project, and serves thousands of educators annually, and partners with school districts across Southern California. The UCI History Project, established in 1999 within the Department of History, has delivered multiple long-term federally funded teacher-development programs for history-social science teachers, including Teaching American History and Improving Teacher Quality programs, and maintains active partnerships with history faculty who provide content expertise aligned to California's frameworks and standards. The California Reading and Literature Project (CRLP), housed in the UCI School of Education, is the UC Irvine/Orange County regional site of one of California's longest-standing Subject Matter Projects. CRLP operates through a robust statewide network of literacy experts, coaches, and educators spanning county offices of education, school districts, and institutions of higher education, giving it the reach and credibility to deliver research-based professional learning at scale across California's diverse regions. Over decades, CRLP has built professional learning that integrates language development with evidence-based literacy instruction, with longstanding expertise in supporting English Learners and the linguistically diverse students who fill California's classrooms. This infrastructure provide an established foundation for developing, delivering, and disseminating the *American Roots* program's curriculum and professional learning.

The UCI Department of History faculty have a long-standing commitment to partnering with UCI History Project staff to support K-12 educators by developing lectures and presentations on history content aligned to California's Frameworks and Standards, support curriculum development by identifying appropriate primary and secondary sources, and engage

in discussions about history methodology and academic literacy instruction. Over 75% of the History Faculty have participated in History Project professional development in this way (See Appendix D for letter).

Through their professional development programs, the *American Roots* team has long-term partnerships and strong networks with educators in local Southern California school districts, collaborating with educators who are committed to implementation of civic engagement in the history and government classrooms. UCI is located in Orange County, which serves a total 437,276 public school students across twenty-eight school districts. The Teacher Academy serves educators in 87% of all the secondary districts in Orange County, thirteen out of fifteen districts. The *American Roots* program will work with partner districts (See letters in Appendix D) to identify 5th grade Teacher Leaders and Teacher Practitioners to participate in our program.

The **Digital Learning Lab** was founded by Dr. Warschauer over a decade ago to use cutting-edge approaches for better understanding students' learning processes, from pre-school to graduate school, as well as to design and develop new tools and resources to improve digital learning opportunities. The development and diffusion of digital media are transforming the terrain of learning in both higher education and K-12 schools. The use of digital media in education is also dramatically expanding the amount and types of data available to analyze teaching and learning processes. The lab has over 15 members working on multiple ongoing research, including:

- AI and Computing for ALL--A collaboration between university researchers and K-12 practitioners to promote computational thinking for multilingual students;

- Generative AI in Education--Initiatives to incorporate generative AI in K-12 and higher education and to promote AI literacy among educators and students; and
- Converse to Learn-- Explores educational applications of AI-based conversational agents, including audio storybooks to promote early literacy skills and science videos to foster scientific thinking.

The Digital Learning Lab has offices to support research and collaboration activities and employs a software engineer, curriculum specialist, and communications specialist to support its activities. The Lab's website (digitallearninglab.org) hosts information on the Lab's research and links to open access resources such as professional development and curriculum. Alumni from the Lab can be found teaching in higher education and working in industry.

The **School of Education** has resources to further its research and teaching mission to foster the success of the *American Roots* program. The School's Business Services Office comprises a financial manager, two contracts and grants analysts, a personnel manager, and two administrative assistants. The office provides projects with grant administration services including budgeting, forecasting, accounting and financial management, and human resource management. On the larger UC Irvine campus, the Office of Research Administration (ORA) consists of Sponsored Projects, Human Research Protections, Research Assurances, Conflict of Interest, and an ORA Training group. ORA is the office of record for extramural proposals and awards supporting research, and education and public service activities of UCI faculty, staff, and students. Staff manage faculty-based regulatory review functions as required by federal and state regulations and UC policy. ORA personnel act as administrative officials with external sponsors, regulatory agencies, higher education organizations, and professional societies regarding

regulatory changes, institutional policy developments, and implementation of regulatory requirements and enhancements. ORA staff members are expert resources for policy and program information. Post-award administration is provided primarily by Contracts & Grants Accounting under the direction of the campus Controller. The Contracts & Grants unit is responsible for financial reporting and administration of extramural funding.

The campus has the capacity to host all of our programming and meetings at the School of Education and the School of Humanities and has the logistical capacity in place to host high-quality professional development gatherings. These meeting locations are all physically accessible locations complying with the Americans with Disability Act requirements. Additional accommodations for teachers with disabilities will be made including materials in accessible formats (General Education Provisional Act Section 427 Assurance). Grant team members have secure individual offices and secure storage space for project records. Finally, it includes two video conference rooms, one seating 5-10 people and one seating 30 people, that can be used for all team meetings.

The *American Roots* program artifacts will be digitized, archived and published on the Teacher Academy's website (https://teacheracademy.uci.edu/) to remain as a resource for educators beyond the scope of the grant period. The grant activities, including six Curricular Units and teaching resources, will continue to be free and readily available online resources for educators in our network and across the US. These grant artifacts will be widely disseminated to educators throughout the US on our websites, including a *American Roots* program website and the statewide networks of the California History-Social Science Project (of which UCI History Project is part) and the California Reading and Literature Project digital communications, which reach over 15,000 educators throughout the US. The teachers impacted by the grant will further

share their learning with teachers in our region through dissemination at their school sites and in the UCI Teacher Academy network, serving as teacher leaders for future programming to foster civic learning for an informed citizenry.

Project costs are reasonable in relation to the number of persons served, the depth and intensity of services, and the anticipated results. Over three years the project provides sustained, year-long professional development to approximately 20 Teacher Leaders and approximately 90 5th grade Teacher Practioners; produces six classroom-ready, openly licensed units with facilitator materials; and delivers a place-based study trip that yields durable curriculum. During the summer of Year 2, Teacher Leaders and Grant Curriculum staff will engage in a study tour on the East Coast (Philadelphia, Washington DC, and Eastern Shore of Maryland), to deepen their understanding of the important historical events and deepen their place-based understanding of the historical context and impact of America's founding on these locations. This experiential learning will create the content knowledge expertise Teacher Leaders and Grant Curriculum staff need to develop the Year 3 units. All six units and materials are released for voluntary national use at no marginal cost to adopters, the project's reach–and thus its cost-effectiveness per ultimate beneficiary–extends well beyond the directly funded participants. The project also leverages substantial non-grant resources: existing district partnerships, History Department faculty effort, the Digital Learning Lab's tools and staff, and campus facilities and administrative infrastructure, concentrating federal funds on the distinctive new work of unit development, professional development, place-based learning, and evaluation.

Project Evaluation

American Roots will be assessed through an internal evaluation led by Co-Principal Investigator Dr. Tamara Tate, integrating formative and summative components aligned to the project's logic model (Appendix A). The evaluation is designed to satisfy the Department's performance objective for the AHC-NA program—that participants demonstrate, through pre- and post-assessments, increased understanding of American history, civics and government, and geography—and to report annually on Program Performance Measure 1, the average percentage gain on that assessment. Because the evaluation is internal and the Co-PI also leads the project's Human-First AI component, the design incorporates explicit objectivity safeguards, described below, that map to the three attributes of quality data specified in 34 CFR 77.1: utility, objectivity, and integrity.

(i) Objective performance measures producing quality quantitative and qualitative data. The evaluation's measures span the logic model's causal chain from professional learning to teacher knowledge, from knowledge to instructional practice, and from practice to student learning, and are aligned to each round's institute content so that assessment tracks instruction.

Teacher content knowledge (confirmatory; Program Performance Measure 1). Working with History Faculty Advisor Dr. Sharon Block, the project will develop three round-specific pre/post assessments of participants' content knowledge, each aligned to that round's founding-principles content—the Declaration of Independence; the Constitution and Bill of Rights; and place-based history and geography—and covering history, civics and government, and geography as the performance objective requires. Instruments will combine selected-response and

constructed-response items and will be developed and reviewed for content validity before the first institute, following the round-aligned assessment approach used successfully in prior UCI History Project programming. The project will report the average percentage gain across participants as its federal performance measure.

Teacher disciplinary-literacy instruction and self-efficacy. The evaluation will assess growth in teachers' capacity to teach reading and writing of primary sources within the CLIC framework, drawing on the measures of teacher knowledge for disciplinary instruction and instructional self-efficacy developed through the IES-funded WRITE Center, on which Dr. Tate served as a researcher. These proximal-practice measures connect the professional learning to changes in instruction.

Unit quality, disciplinary accuracy, and teacher-facing AI integration. A program-designed rubric will score the inquiry units teachers produce for content accuracy, alignment to the California History–Social Science Framework and the CLIC shifts, and quality of inquiry and source use. Because generative AI can introduce historical inaccuracies, the rubric will include an explicit disciplinary-accuracy verification component that documents whether teachers applied the project's source-verification protocol when using AI to build, modify, or extend curriculum; this doubles as a measure of the AI component's central design claim—that AI augments rather than displaces teachers' disciplinary judgment. Teachers' AI-supported instructional planning may additionally be assessed using an adapted AI-TPACK or AI-knowledge instrument [candidate instruments Ning et al., 2024; Islam & Mishra, 2024].

Student outcomes (pilot classrooms). In the four Curriculum Pilot Teachers' fifth-grade classrooms each round, the project will collect pre/post measures of students' content understanding and primary-source analysis, and will track participation in the K–8 Civic Learning Pathway toward the State Seal of Civic Engagement as a distal civic-learning indicator.

Qualitative data. The evaluation will generate qualitative evidence through the structured review of teacher-created lessons using the project's continuous-improvement protocol, open-ended survey items, interviews or focus groups with teacher leaders, and analysis of how teachers revise AI-generated materials—capturing dimensions of disciplinary reasoning that rubric scores alone do not.

To secure the objectivity and integrity of these data given the internal design, instruments and scoring rubrics will be specified in advance; artifact scoring and student-assessment coding will be conducted by the Graduate Student Researcher or a neutral research assistant rather than by component leads, with scorers blind to condition where a comparison is involved; and a written analysis plan will be registered before outcome data are analyzed.

(ii) Evaluation design, evidence of effectiveness, and information to guide replication. Analyses will estimate pre/post change with baseline scores as covariates and will use hierarchical linear models for student outcomes nested within classrooms, reporting effect sizes alongside gains and monitoring attrition and baseline equivalence between comparison groups. Formatively, the evaluator will deliver a report at each round's key decision points, synthesizing survey data, lesson reviews, and assessment results into specific recommendations that feed a Plan–Do–Study–Act revision cycle before the subsequent round—so that each cohort benefits

from evidence gathered in the one before it. To support replication, the project will document the fidelity of institute delivery and unit implementation and will release openly licensed units, an implementation guide, and archived instruments and analytic code.

The team's capacity to collect and report reliable, valid, and meaningful data is established by Dr. Tate's methodological expertise (development of teacher-knowledge and self-efficacy measures and hierarchical modeling of instruction–achievement relationships through the WRITE Center, and four grants advancing AI-supported teaching and learning), Dr. Block's experience developing content assessments, and the UCI History Project's record of externally evaluated professional development, with data support from the Graduate Student Researcher.

Goal 1: Increase teacher content knowledge of founding principles and documents. All Teacher Leaders and Teacher Practitioners attending the summer institute will be assessed on the impact of the grant on their content knowledge. Dr. Gilbertson and Dr. Tate will lead the development of both quantitative and qualitative assessments to understand the grant impact. The quantitative assessment will be developed to directly align with the content topics and presentations to the teachers by Dr. Block and the historians who will present to the teachers during the academic year and in the summer institute. For each round, the team administers a validated content-knowledge assessment to Teacher Leaders and Teacher Pracitioners before and after the summer institute. Assessment items, multiple-choice and short-constructed-response, will center on the founding documents and to the units taught, developed during instrument design and reviewed for content validity. The primary quantitative outcome is the average percentage gain from pre- to post-assessment, reported annually as the federal performance

measure (Program Performance Measure 1). In addition, teachers will be assessed qualitatively through written surveys where teachers reflect on the impact of the historian's presentations. This will allow them time to process the information as it relates to the development of their content knowledge and how this information will be utilized in the teaching unit.

Goal 2: Build teachers' capacity to use AI as a pedagogically guided support for designing, adapting, and refining curriculum and instructional materials on the founding documents. The evaluation will assess not only the extent to which teachers use PapyrusAI, but also the quality of that use and its contribution to curriculum development, instructional practice, and student learning. The team will analyze logs of teacher-AI interactions to document patterns of use, including frequency of access, number and type of prompts, conversational turns, revision cycles, and the instructional purposes for which teachers use the tool. These quantitative indicators will show whether and how teachers engage with PapyrusAI across the curriculum development and implementation cycle.

Qualitative analysis of teacher-AI interaction logs will examine evidence of Human-First AI use, including the extent to which teachers provide strong instructional context, use iterative prompting, question or revise AI outputs, verify historical claims, and adapt materials based on students' needs and classroom context. The team will also code for pedagogically sound AI use, with attention to content accuracy, primary source integration, inquiry design, disciplinary literacy supports, scaffolding, assessment alignment, and evidence that the teacher remains the author, evaluator, and instructional decision-maker.

To assess whether AI use contributes to stronger curriculum, the evaluation team will review AI-supported curriculum revisions produced by Teacher Leaders and Teacher Practitioners each year. Curriculum artifacts will be assessed with a review rubric focused on

historical accuracy, use of primary sources, alignment to founding documents and principles, inquiry design, disciplinary literacy supports, accessibility for fifth-grade learners, and evidence of teacher reflection and revision. This analysis will allow the team to distinguish superficial AI use from substantive, pedagogically guided use that improves the quality and usability of instructional materials.

Teacher Leaders and Teacher Practitioners will also maintain year-long digital journals documenting how they use AI, what they learn from the process, how they evaluate and adapt AI-generated materials, and how AI-supported planning affects their instruction. These journals will serve both as a professional learning tool and as a qualitative data source. Because "Reflect and Learn" is the final step of the Digital Learning Lab's Human-First AI Framework, the journals will provide evidence of teachers' developing capacity to use AI thoughtfully, critically, and in ways that preserve professional judgment.

Finally, the evaluation will examine whether AI-supported curriculum development is associated with the student learning outcomes identified in the logic model. Student work from implemented units will be analyzed using a team-developed rubric focused on students' ability to demonstrate understanding of founding principles, use evidence from primary sources, construct historically grounded explanations, and explain the significance of those principles in American civic life. Together, the AI interaction logs, curriculum artifacts, teacher journals, implementation reflections, and student work samples will provide a triangulated account of how pedagogically guided AI use supports teacher capacity, curriculum quality, and student learning.

Goal 3 Develop 6 inquiry-based, primary source units to support history, civics, and geography in 5th grade classrooms. Each unit, the central deliverable, is scored against an explicit unit-review rubric assessing alignment to the California History–Social Science

Framework, primary-source rigor, disciplinary-literacy supports, inquiry design, and citizenship focus (the CLIC criteria). Units must meet a quality threshold before dissemination, and rubric scores document quality objectively. The writing, implementation, and revision process will allow for the improvement of the units to meet the specific needs of teachers and their students in our partner districts. Teacher Curriculum Writers will work with historians and the grant team to write the units in the winter, Curriculum Pilot Teachers, will pilot the lessons in the winter and spring using student data and their own reflection to provide feedback for the grant team who will then revise the units to share with the summer institute teachers. At each step in the process, the unit will be thoroughly evaluated to best meet the needs of 5th grade students and aligned to the rubric criteria.

Teacher Practitioner Implementation during the academic year is measured through teacher self-report on a structured instrument and through instructional artifacts, such as lessons taught, sources used, and student work samples. Student learning in content knowledge and disciplinary literacy is assessed through student work products scored with rubrics co-developed by the team and Teacher Leaders and aligned to State content and literacy standards. AI-related outcomes are measured through teacher-improved assignments and assessments and through reflections coded for application of the Human-First AI Framework, including the use of AI to analyze student work and identify next steps.

Goal 4: Build a sustainable Teacher Leader network that supports rigorous inquiry-based instruction in the founding documents Teacher Leaders and Teacher Practitioners will be assessed quantitatively and qualitatively to understand their efficacy in integrating the high-quality units in their classrooms to meet student learning needs. The PI and grant team will develop a survey instrument to assess collective efficacy using validated assessment survey

questions (Voelkel and Chrispeels, 2017). This will be further studied through teacher self-reports of implementation and qualitative survey questions that allow teachers to demonstrate their own understanding of the implementation of the units for their classroom context. A particular focus of the efficacy assessment will be to understand teachers efficacy in teaching the founding documents to support student learning. The focus on student work and student work analysis as a learning community will be assessed also through our professional development meetings and focus groups interviews.

To evaluate progress toward Goal 4, the project will assess the extent to which *American Roots* builds a sustainable teacher-leader network that supports rigorous, inquiry-based instruction on the founding documents in fifth-grade classrooms. The evaluation will examine both implementation and capacity-building, whether Teacher Leaders and Teacher Practitioners integrate the units into regular instruction, participate in reflective discussions about implementation, use student work to inform instructional decisions, and develop greater individual and collective efficacy for teaching history, civics, and geography.

Quantitative data will include teacher implementation reports, participation records for professional learning and follow-up meetings, and pre/post survey measures of teacher efficacy and collective efficacy. The evaluation will assess both individual teacher self-efficacy and collective teacher efficacy. Teacher self-efficacy refers to teachers' confidence in their own ability to teach the founding documents, support students' close reading of primary sources, facilitate evidence-based discussion, and adapt inquiry-based units for fifth-grade learners. Collective efficacy refers to the shared belief among Teacher Leaders and Teacher Practitioners that, as a professional learning network, they can support rigorous history, civics, and geography

instruction and improve student learning. Survey items will draw on validated teacher efficacy and collective efficacy measures and will be adapted to the specific context of inquiry-based instruction on the founding documents (Goddard et al., 2000; Tschannen-Moran & Woolfolk Hoy, 2001; Voelkel & Chrispeels, 2017).

Qualitative data will include focus group interviews, open-ended survey responses, teacher reflections, implementation logs, and records from professional development meetings. These data will be analyzed to understand how teachers describe their implementation of the units, the instructional adaptations they make for their students, the challenges they encounter, and the ways participation in the teacher-leader network supports continued improvement. A particular focus will be placed on how teachers use student work analysis as a shared professional practice to identify evidence of student learning, discuss instructional next steps, and refine curriculum and instruction in response.

Together, these data will provide evidence of the degree to which *American Roots* strengthens teachers' capacity to implement rigorous history, civics, and geography instruction and whether the project is building the conditions for sustainability beyond the grant period. Evidence of success will include increased teacher efficacy, regular implementation of the units, productive use of student work in professional learning, stronger collaboration among Teacher Leaders and Teacher Practitioners, and teacher-reported plans to continue using and adapting the units in future years.

Program evaluation will be led by Dr. Gilbertson and Dr. Tate. Dr. Gilbertson has led previous federal teaching grant programs for history educators including Teaching American

History grant programs, Improving Teacher Quality Grant programs, and a National Activities Grant program. These programs resulted in improved student outcomes. A partnership between Santa Ana Unified School District and the UCI History Project, evaluated by Silvia Swigert, Director of Evaluation for the Center for Educational Partnerships at UCI, demonstrated students of teachers involved in the programs increased their knowledge of US history content, writing skills, and primary source analysis. Similarly, a 2009 evaluation of Orange Unified School District's partnership with UCI, conducted by Continuous Improvement Associates, found that English Learners were positively affected by their teachers' participation in the Literacy in the History Classroom program. These programs demonstrated a positive impact on teacher and student learning.The data collection instruments will be developed using validated assessment surveys and also utilize assessment questions that are directly aligned to the professional learning of the *American Roots* program to understand the specific impact of the grant program on participating teachers.

Evidence Building The *American Roots* program will gather quantitative and qualitative data to yield information that guides replication and to produce valid and reliable information about the effectiveness of the approach. The evaluation gathers qualitative data through analysis of student work, teacher reflections, and structured review of teacher-authored units, and collects formative feedback after each professional-development activity. Quantitative data analysis will support understanding of teacher content knowledge improvement and teacher efficacy. Consistent with the project's continuous-improvement process, findings are reviewed on a regular cycle and used to refine instruments, institutes, and units in each subsequent round. The evaluation is designed to yield information that guides replication and to produce valid and reliable information about the effectiveness of the approach. Under the program's evidence framework, the project

demonstrates a rationale for its central component and includes ongoing efforts to examine its effects. To support replication, the project documents its full model: the unit template and the six completed units, facilitator guides, the pre/post assessments and scoring rubrics, the implementation instrument, and a program-implementation summary describing the teacher-leader cycle, the place-based design, and the human-first AI approach. All such materials are released openly for voluntary national use.

To strengthen inference about effectiveness within the project's scale, the design uses pre/post comparisons on validated measures for every cohort and examines whether variation in implementation is associated with variation in student-work quality. Where feasible, the staggered round structure permits descriptive comparison between teachers who have and have not yet received the professional development, and the pilot-to-cohort sequence allows the team to document how units change as evidence accumulates. The team will be transparent about the limits of these comparisons and report results so that other educators and researchers can judge transferability to their own settings.

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